Sensitivity via Omitted Variable Bias

PS200C, Spring 2025

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Outline

1. Background
2. Example and Motivation: Darfur
3. Sensitivity analysis with omitted variable bias
4. Tools

Learning sensitivity analysis

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- gives you a lens through which to view existing work beyond just complaining about non-zero confounding

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- *Wildfire Exposure Increases Pro-Climate Political Behaviors*
- Demos at http://carloscinelli.com/sensemakr/articles/

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What is the average effect of exposure to violence on attitudes toward peace (more "angry" or more "weary"), among Darfurian refugees in eastern Chad?

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- Purpose of violence was generally to remove the population rather than kill those likely to be rebels
- Bombing from the air: extremely crude, difficult to target, and no information
- Attacks by *Janjaweed* militia: also highly indiscriminate, had only visual characteristics on which to target (notably, *female*)

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But what if the ID strategy is imperfect?

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In political science,

- Only 6% of papers involving a causal ID strategy other than experiments conducted a formal analysis of sensitivity to unobserved confounders (covering APSR, AJPS, JOP in 2017 and in 2019).

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5. What if we have a weak identification strategy – can we say anything about what effect might survive a conservative assumption about the amount of confounding?

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2. Contour plots exploring how point estimate, t-statistic, CI would change depending on hypothesized degree of confounding
3. An approach that transforms assumptions about the strength of observables vs. confounders into bounds on the allowable amount of confounding.
4. Analysis of “extreme scenarios” when investigators are in a weak position to argue for limited confounding.

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Whether that full model is a wise one to run or corresponds to the quantity of interest up to you!

Omitted Variable Bias

By "partialling out", we can use the traditional OVB formulation on multivariate data:

$$\begin{aligned}\hat{\tau}_{\text{res}} &= \frac{\text{cov}(D^{\perp \boldsymbol{X}},\ Y^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})} \\ &= \frac{\text{cov}(D^{\perp \boldsymbol{X}},\ \hat{\tau} D^{\perp \boldsymbol{X}} + \hat{\gamma} Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})} \\ &= \hat{\tau} + \hat{\gamma}\hat{\delta} \end{aligned} \tag{3}$$

where $Y^{\perp \boldsymbol{X}}$, $D^{\perp \boldsymbol{X}}$ and $Z^{\perp \boldsymbol{X}}$ are the variables Y, D and Z after removing the components linearly explained by $\boldsymbol{X}$ and we define $\hat{\delta} := \frac{\text{cov}(D^{\perp \boldsymbol{X}}, Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}$.

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Note that $\hat{\delta}$ is the coefficient you would get if you could run:

$$Z = \hat{\delta} D + \boldsymbol{X}\hat{\psi} + \hat{\epsilon}_Z \tag{4}$$

OVB continued

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- Key lesson: computing the consequence of omitting Z on the estimate of τ requires knowing *only* two things:
  - how Z relates to Y ("impact", γ)
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- Key lesson: computing the consequence of omitting Z on the estimate of τ requires knowing *only* two things:
  - how Z relates to Y ("impact", γ)
  - how Z relates to D ("imbalance", δ)
- This gives basis for a simple sensitivity analysis without assumptions about distribution of τ, treatment assignment, etc.

Darfur example: sensitivity with OVB

Suppose a colleague argues:

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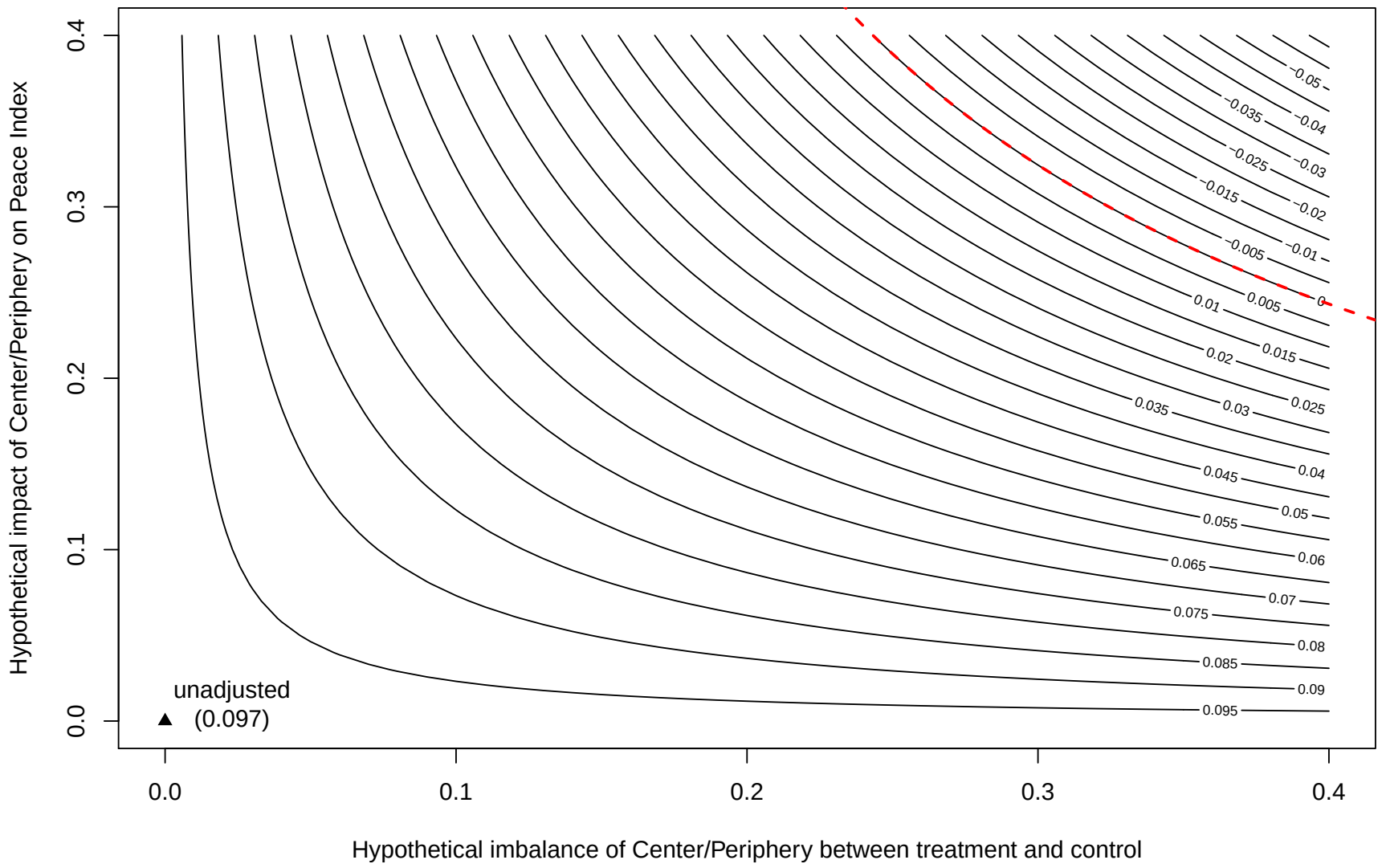
- How bad would this *Center* confounder have to be to be problematic?
- Consider a contour plot where, at each pair of hypothesized "impact" and "imbalance" compute adjusted effect estimate, i.e $\hat{\tau} = \hat{\tau}_{\text{res}} - \hat{\gamma}\hat{\delta}$


Hypothetical impact of Center/Periphery on Peace Index
0.0
0.1
0.2
0.3
0.4
unadjusted
(0.097)
0.095
0.09
0.085
0.08
0.075
0.07
0.065
0.06
0.055
0.05
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
−0.005
−0.01
−0.015
−0.02
−0.025
−0.03
−0.035
−0.04
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Hypothetical imbalance of Center/Periphery between treatment and control

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We reparameterize "impact" and "imbalance" using partial R^2 (of Z with D and of Z with Y)

- gives scale-free yet intuitive measure of association
- turns out to make a number of other tasks possible or easier: multiple Z, non-linear action, bounding, variance

Bias formulas

$$|\widehat{\text{bias}}| = \text{se}(\hat{\tau}_{\text{res}})\sqrt{\frac{R^2_{Y\sim Z|\boldsymbol{X},D}\, R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}(\text{df})} \tag{5}$$

$$\text{se}(\hat{\tau}) = \text{se}(\hat{\tau}_{\text{res}})\sqrt{\frac{1 - R^2_{Y\sim Z|\boldsymbol{X},D}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}\left(\frac{\text{df}}{\text{df} - 1}\right)} \tag{6}$$

Backup: what is the partial R^2

First, the partial correlation, is the correlation after partialing out, e.g. $R_{Y \sim D|X} = cor(Y^{\perp X}, D^{\perp X})$. Square that to get the partial R^2.

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$R^2_{D \sim Z|X}$, for example can be thought of as the "proportion of variance explained in D by Z, after accounting for X. Specifically, we mean:

$$R^2_{D \sim Z|\boldsymbol{X}} = \frac{R^2_{D \sim \boldsymbol{X}+Z} - R^2_{D \sim \boldsymbol{X}}}{1 - R^2_{D \sim \boldsymbol{X}}} \tag{7}$$

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Similarly you could say:

$$R^2_{Y \sim Z|D,\boldsymbol{X}} = \frac{R^2_{Y \sim D+\boldsymbol{X}+Z} - R^2_{Y \sim D+\boldsymbol{X}}}{1 - R^2_{Y \sim D+\boldsymbol{X}}}, \tag{8}$$

Note that the direction does not matter:
$R^2_{Z \sim D} = \text{cor}(Z, D)^2 = \text{cor}(D, Z)^2 = R^2_{D \sim Z}$.

Deriving the bias formula in the R^2 parameterization

By FWL,

$$
\begin{aligned}
\widehat{\text{bias}} &= \hat{\delta}\hat{\gamma} \\
&= \left(\frac{\text{cov}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}\right)\left(\frac{\text{cov}(Y^{\perp \boldsymbol{X},D},\ Z^{\perp \boldsymbol{X},D})}{\text{var}(Z^{\perp \boldsymbol{X},D})}\right) \\
&= \left(\frac{\text{cor}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})\text{sd}(Z^{\perp X})}{\text{sd}(D^{\perp X})}\right)\left(\frac{\text{cor}(Y^{\perp X,D},\ Z^{\perp X,D})\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(Z^{\perp \boldsymbol{X},D})}\right) \\
&= \left(\frac{\text{cor}(Y^{\perp \boldsymbol{X},D},\ Z^{\perp \boldsymbol{X},D})\text{cor}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})}{\frac{\text{sd}(Z^{\perp \boldsymbol{X},D})}{\text{sd}(Z^{\perp \boldsymbol{X}})}}\right)\left(\frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})}\right) \qquad (9)
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Noting that $\text{cor}(Y^{\perp \boldsymbol{X},D}, Z^{\perp \boldsymbol{X},D})^2 = R^2_{Y\sim Z|\boldsymbol{X},D}$, that $\text{cor}(Z^{\perp \boldsymbol{X}},\ D^{\perp \boldsymbol{X}})^2 = R^2_{D\sim Z|\boldsymbol{X}}$, and that $\frac{\text{var}(Z^{\perp \boldsymbol{X},D})}{\text{var}(Z^{\perp \boldsymbol{X}})} = 1 - R^2_{Z\sim D|\boldsymbol{X}} = 1 - R^2_{D\sim Z|\boldsymbol{X}}$, we can write 9 as simply,

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$$|\widehat{\text{bias}}| = \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\ R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}}\left(\frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})}\right) \qquad (10)$$

Bias anatomy

$$\text{relative bias} = \frac{\overbrace{|R_{Y\sim Z|\boldsymbol{X},D} \times f_{D\sim Z|\boldsymbol{X}}|}^{\text{bias factor}}}{\underbrace{|f_{Y\sim D|\boldsymbol{X}}|}_{\text{partial f of D with Y}}} = \frac{BF}{|f_{Y\sim D|\boldsymbol{X}}|} \tag{11}$$

- f is from the (partial) Cohen's f^2, which is $R^2/(1-R^2)$.

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- Size of relative bias is the BF compared to $f_{Y\sim D|\boldsymbol{X}}$.
  - $f_{Y\sim D|\boldsymbol{X}}$ is just a transformation of $R_{Y\sim D|X}$. (Also simply $t_D/\sqrt{df}$)
  - $f_{Y\sim D|\boldsymbol{X}}$ is the *only* property of the data that enters.

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- f is from the (partial) Cohen's f^2, which is $R^2/(1-R^2)$.
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  - $f_{Y\sim D|\boldsymbol{X}}$ is just a transformation of $R_{Y\sim D|X}$. (Also simply $t_D/\sqrt{df}$)
  - $f_{Y\sim D|\boldsymbol{X}}$ is the *only* property of the data that enters.
- This form reminds us that sensitivity is an identification problem, unrelated to sample size.

Variance anatomy

$$\text{relative variance} = \overbrace{\left(1 - R^2_{Y \sim Z|\boldsymbol{X},D}\right)}^{\text{VRF}} \underbrace{\left(\frac{1}{1 - R^2_{D \sim Z|\boldsymbol{X}}}\right)}_{\text{VIF}} \overbrace{\left(\frac{\text{df}}{\text{df} - 1}\right)}^{\text{change in df}}$$

$$= \text{VRF} \times \text{VIF} \times \text{change in df} \tag{12}$$

Outline

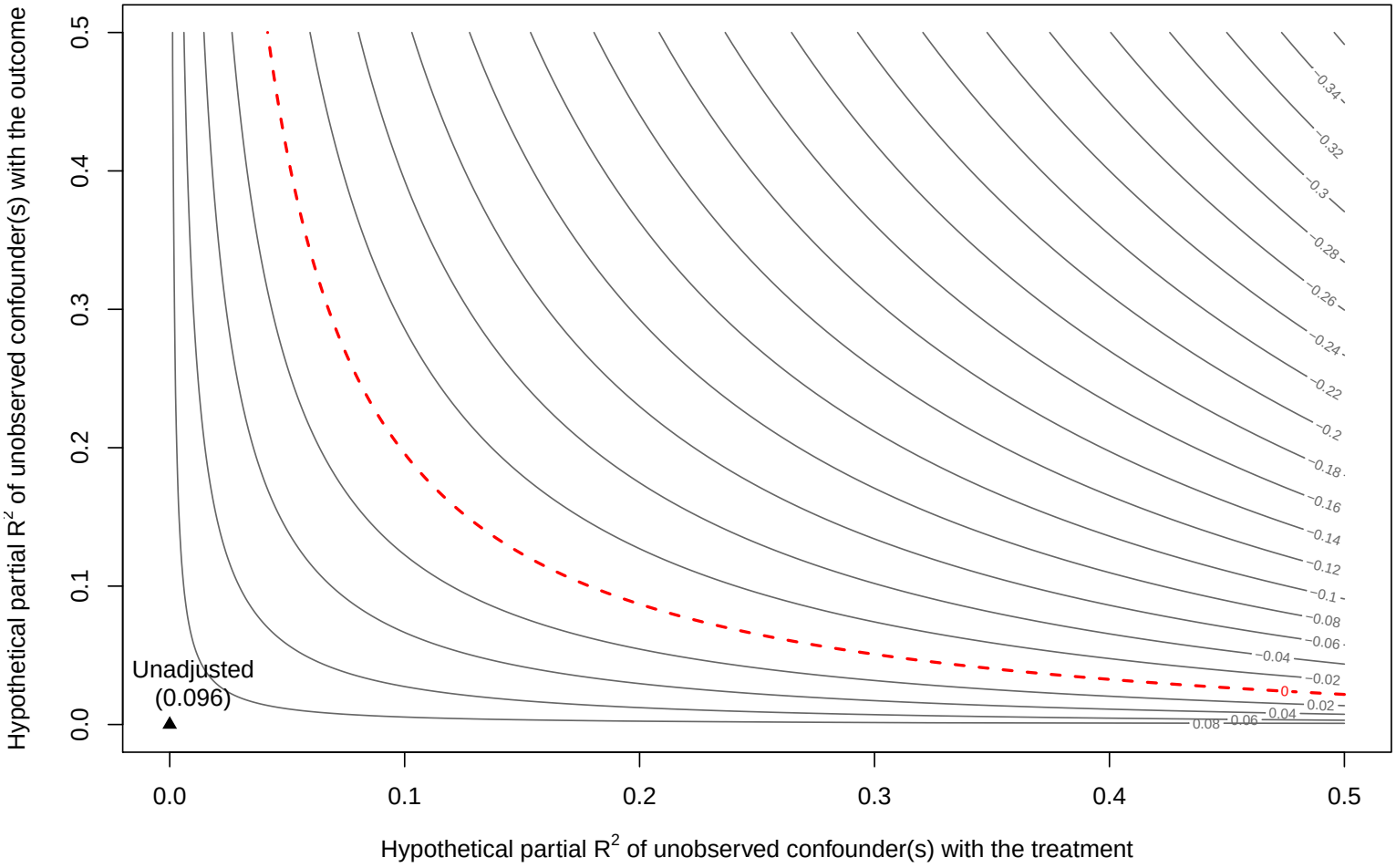
1. Background
2. Example and Motivation: Darfur
3. Sensitivity analysis with omitted variable bias
4. Tools

Tools

1. Two simple statistics for routine reporting: RV and $R^2_{Y \sim D|\boldsymbol{X}}$
2. Contour plots of various confounding-adjusted quantitites, with bounds
3. Extreme-scenario plots when large proportion of residual may be due to confounder.

First, good to have this in your head:

Figure: Contour plot with partial R^2



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- $R^2_{Y \sim D|\boldsymbol{X}}$: partial variance explained by treatment
  - if confounder explained *all* residual variance of the outcome, to bring the estimated effect down to zero, this would equal the partial R^2 of the confounder with the treatment.

Augmented Regression Table: Darfur example

Outcome: *Peace Index*

Treatment:	Est.	SE	t-value	$R^2_{Y\sim D\mid \boldsymbol{X}}$	RV	$\text{RV}_{\alpha=0.05}$
Directly Harmed	0.097	0.023	4.18	2.2%	13.9%	7.6%
df = 783				*Bound (1 × female):* R^2_y = 12%, R^2_d = 1%		

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Somebody tell us:

- what does this RV mean?
- what does thie $RV_{\alpha=0.05}$ mean?
- what does this $R^2_{Y\sim D\mid X}$ mean?

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- They are intended to help "calibrate our intuition" or tell us how a confounder "not unlike X_j" would change our answer.
- But, the bias caused by the confounder changes these observed statistics in problematic ways (even when $\boldsymbol{X} \perp Z$).

Confounding of informal benchmarks

Consider how the coefficient on $\boldsymbol{X}$ is altered by omissions of Z, using the same OVB approach:

$$\hat{\beta}_{\text{res}} - \hat{\beta} = \hat{\gamma}\hat{\psi} \tag{13}$$

where $\hat{\psi}$ is obtained from the regression $Z = \hat{\delta}D + \boldsymbol{X}\hat{\psi} + \hat{\epsilon}_Z$.

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- Thus, claims comparing $\hat{\beta}_{\text{res}}$ with a hypothesized confounder, as if $\hat{\beta}_{\text{res}} \approx \hat{\beta}$ can be very misleading.

Illustrating the benchmark problem

Suppose:

$$X = \epsilon_x \tag{14}$$

$$Z = \epsilon_z \tag{15}$$

$$D = X + Z + \epsilon_d \tag{16}$$

$$Y = X + Z + \epsilon_y \tag{17}$$

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We would thus like a benchmarking analysis to tell us that an omitted confounder "not unlike X" would indeed eliminate our estimated effect of D.

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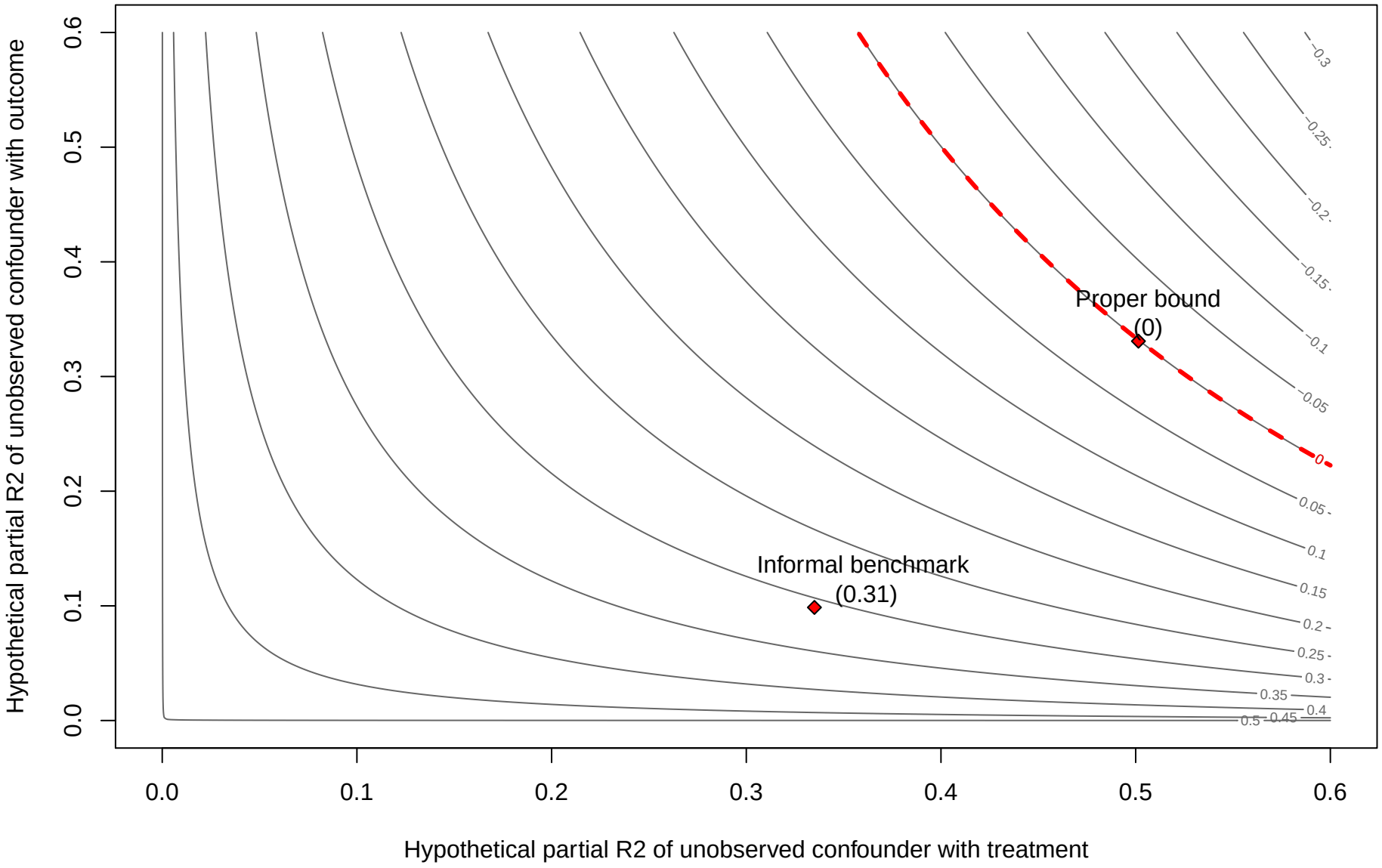

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- Can be done with any sub-group or with all covariates jointly

Result: bounding the confounding subject to assumption

Given some $\{X_j, k_D, k_Y\}$, the strength of the confounders is bounded at:

$$R^2_{D\sim Z|\boldsymbol{X}} = k_D f^2_{D\sim X_j|\boldsymbol{X}_{-j}}, \qquad R^2_{Y\sim Z|\boldsymbol{X},D} \leq \eta^2 f^2_{Y\sim X_j|\boldsymbol{X}_{-j},D} \tag{19}$$

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Allows us to investigate the maximum effect a confounder at most "k times" as strong as a particular covariate X_j would have on the coefficient estimate.

- Certain values for k_D and k_Y may be ruled out by the data (for instance, if $R^2_{D\sim X_j|\boldsymbol{X}_{-j}} = 50\%$ then k_D must be less than 1).

Confounder bounding continued

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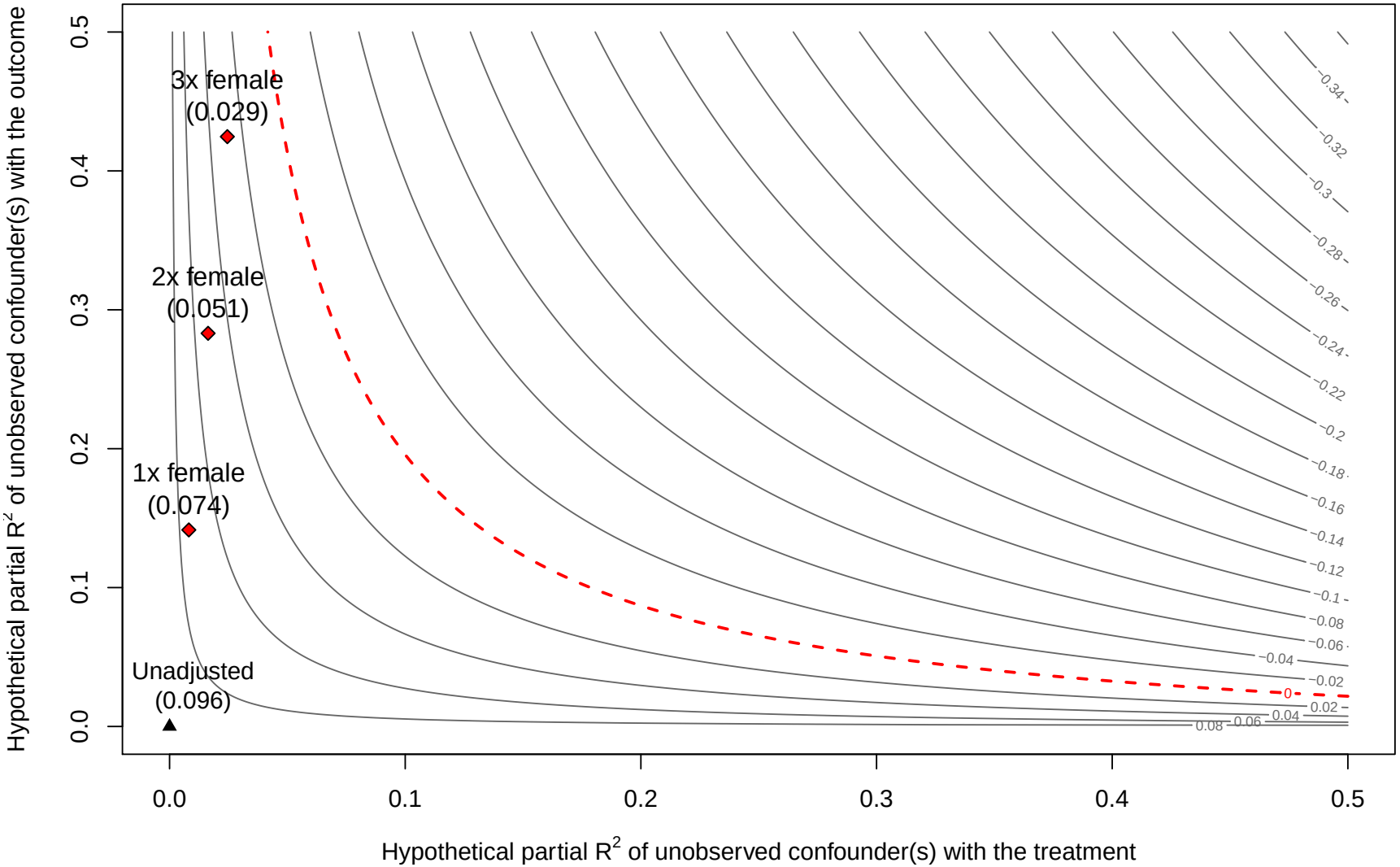
- In a natural experiment where you know about treatment assignment or the outcome, you may have more traction.
- Example: in the Darfur case, within village it is hard to imagine any basis for targeting of violence that could approach the importance of *female* indicator, so let's use it for this assumption.

Contour plots with bounds



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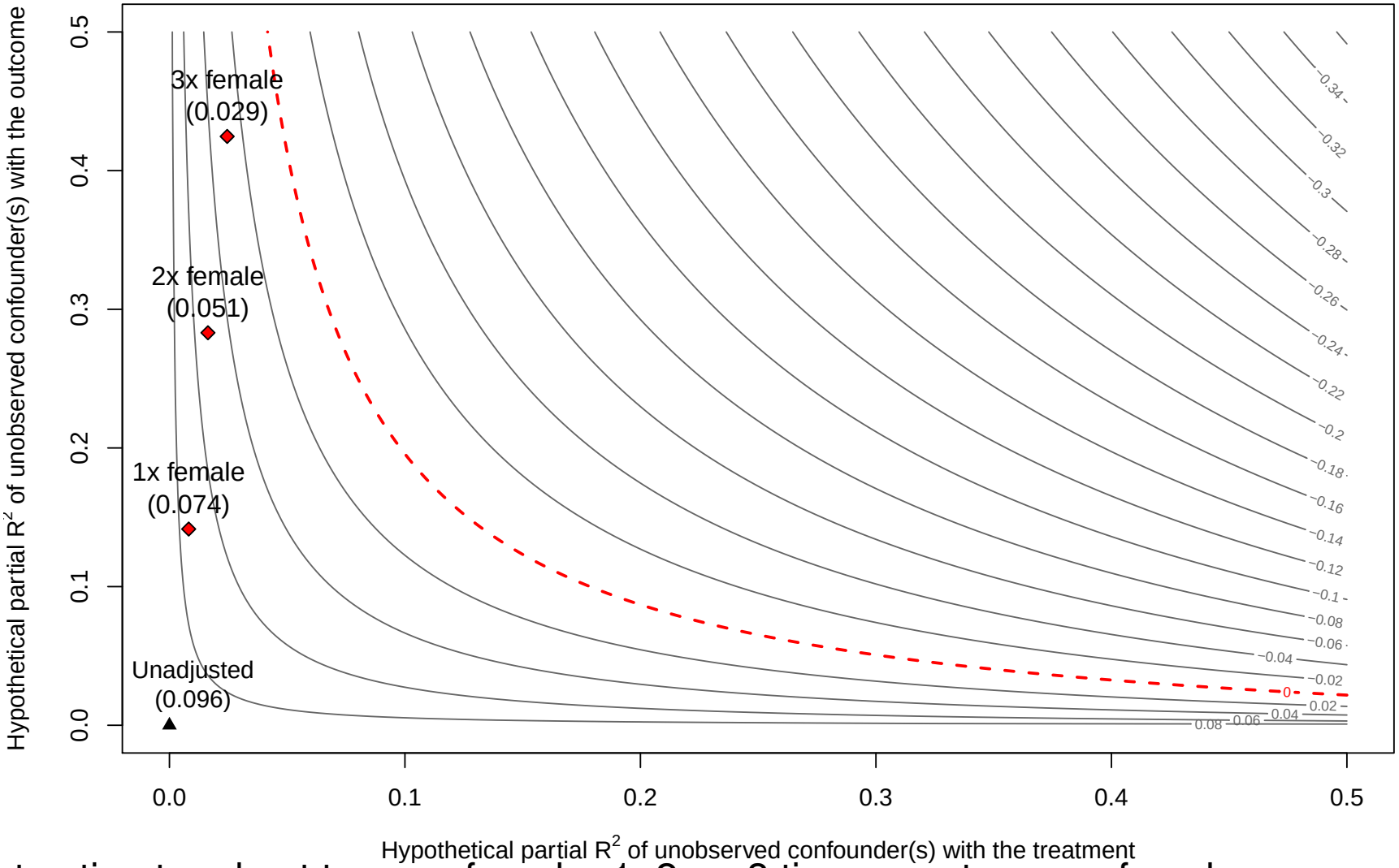
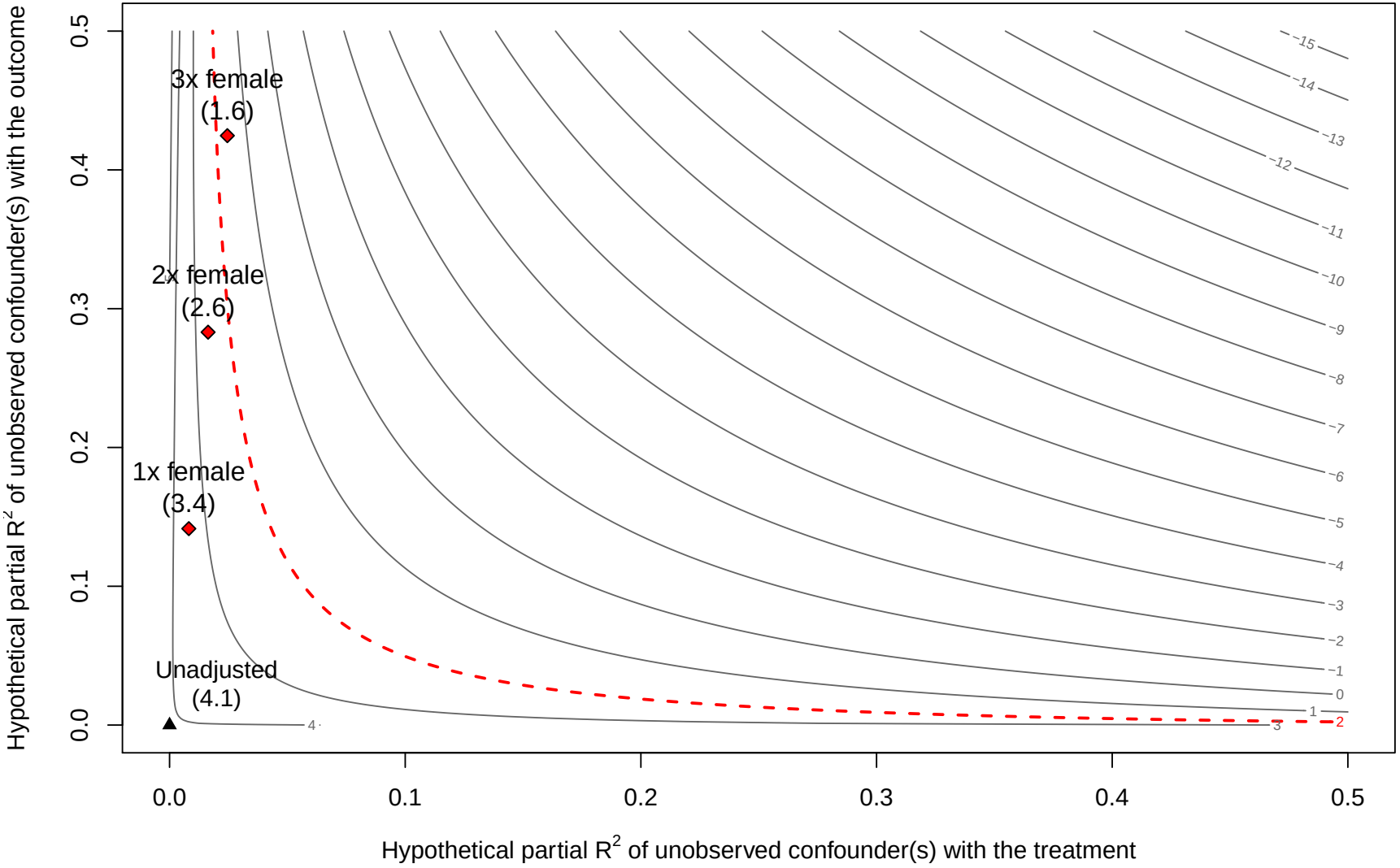


Point estimate robust to a confounder 1, 2, or 3 times as strong as female.
(Conservative in case of multiple, non-linear confounders)

Contour plot of t-statistics



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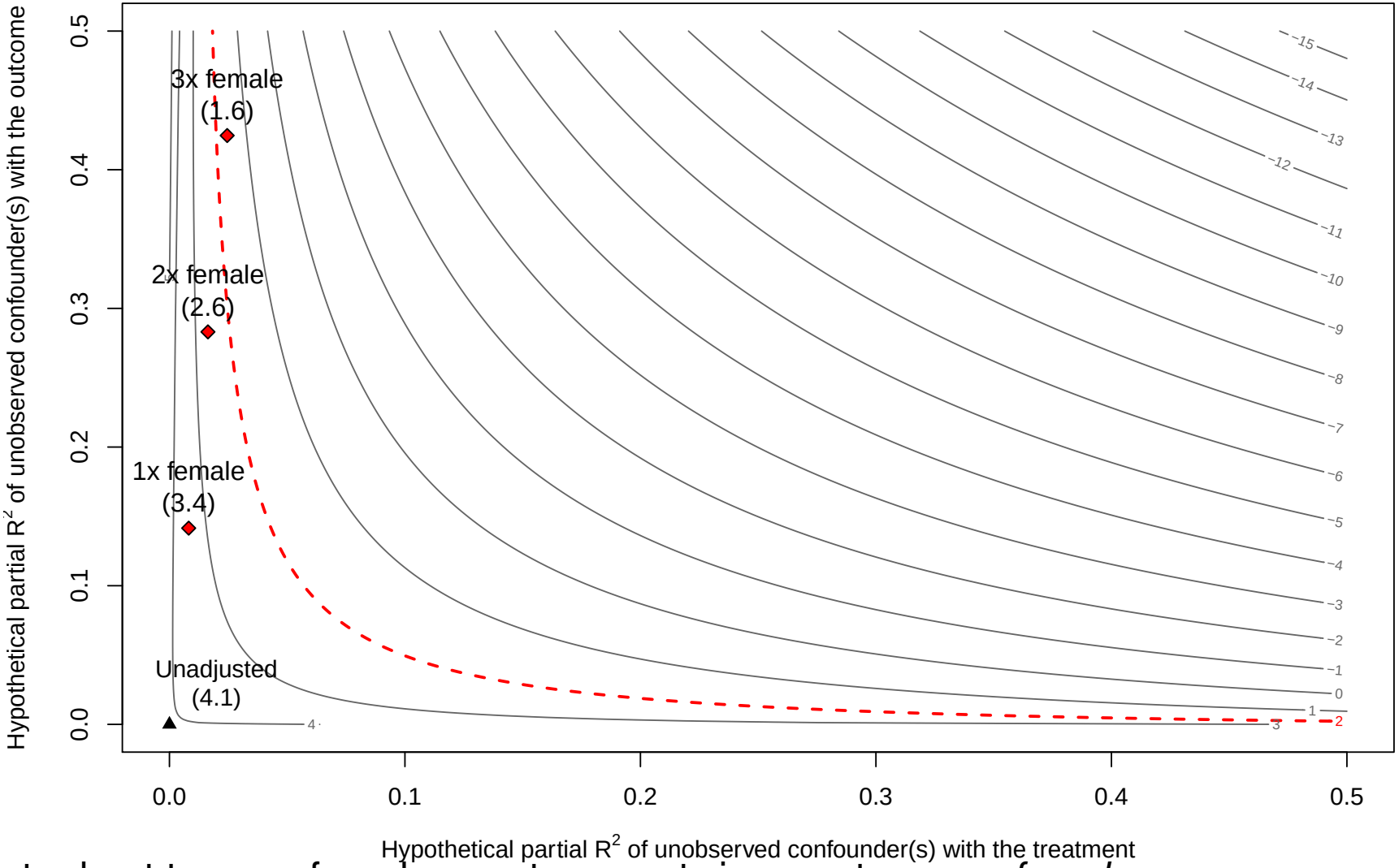

T-stat robust to a confounder as strong or twice as strong as *female*

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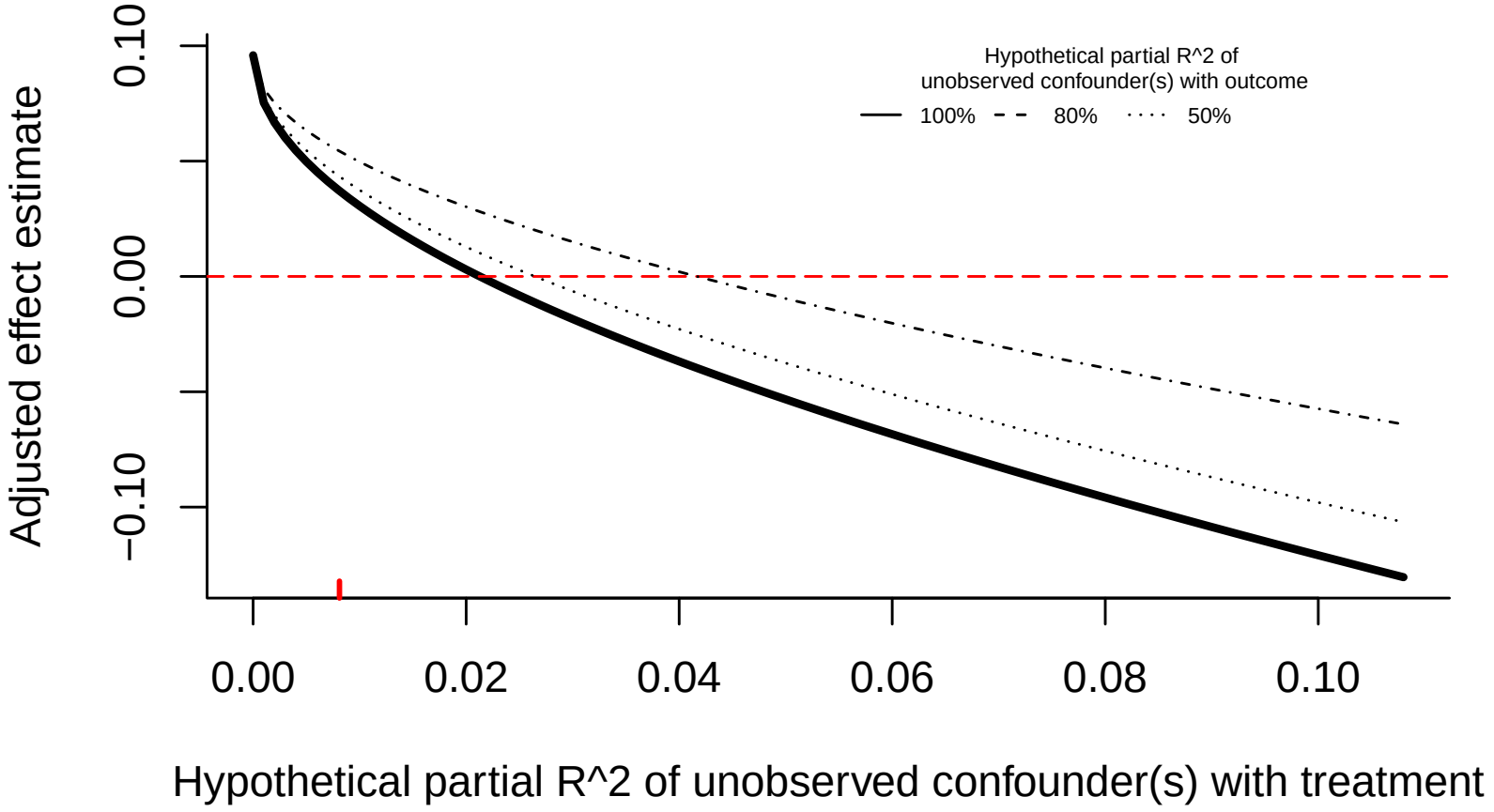
- Often gives bad news: you pay a price for not being able to rule out moderate confounders by design or special knowledge!

Extreme Scenario Plots: Example



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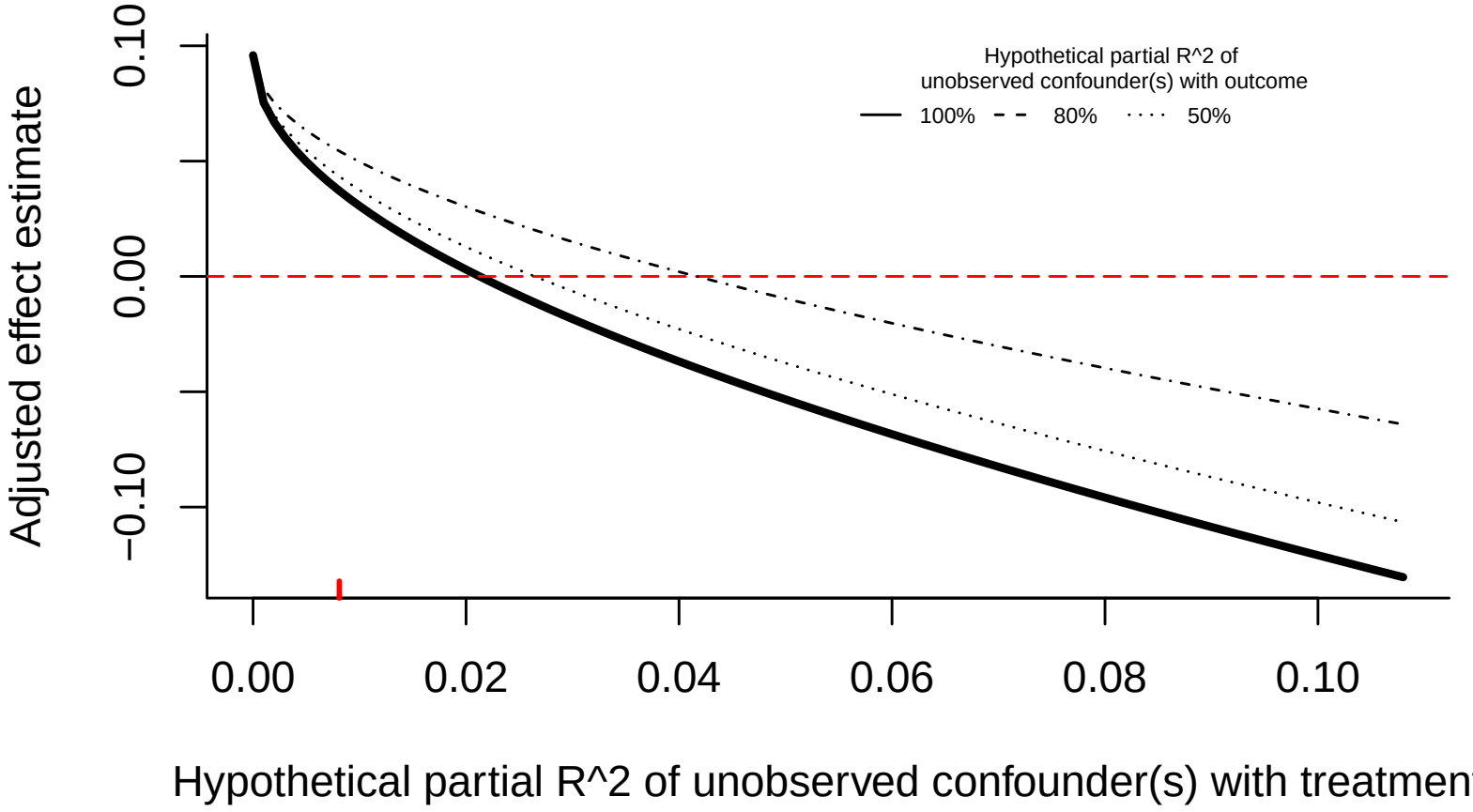

In the most extreme scenario, confounder(s) have to explain 2.2% of the treatment to bring effect to zero (which is exactly $R^2_{Y \sim D|\boldsymbol{X}}$). Compare to bound obtained from "1 x female". (red marker)

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- Extreme scenario analysis does not excuse poor design – you pay a cost for inability to rule out confounders.

Arguing (from conclusion of the software paper)

1. A causal interpretation of the research conclusion may be defended by claiming that, given the way injuries (the “treatment”) occurred, the scope for targeting particular types of individuals was quite limited... aircraft dropped makeshift and unguided bombs and other objects over villages, and militia raided without concern for who they would attack—the only known major exception to this, due to sexual assaults, was targeting gender, which is also one of the most visually apparent characteristics of an individual. Thus, a confounder twice as strong as female would be surprising.

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This is the kind of argumentation I expect when using benchmarks.

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Bigger picture: Moving away from "is it identified", towards more informative debate over "how much confounding would change the conclusion".